Threshold × k | Accuracy and teacher token share

Qwen3-1.7B student / Qwen3-14B teacher · 8 benchmarks × 25 configurations × 2 metrics = 400 values

MATH500

Baseline accuracy: Student 71.60% | Teacher 84.60%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	81.20	82.00	**83.80**	82.20	81.00
2.5	80.20	80.80	81.20	82.00	80.20
3	80.00	79.60	80.60	79.60	78.60
3.5	76.20	79.00	78.80	79.60	80.00
4	77.60	78.20	76.60	77.40	77.80

Color scale: 76.20 – 83.80

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	1.15	4.20	7.63	13.52	23.83
2.5	0.76	2.83	4.95	9.42	17.10
3	0.52	1.86	3.62	6.81	11.93
3.5	0.35	1.29	2.69	5.02	9.15
4	0.26	0.95	1.89	3.46	6.83

Color scale: 0.00 – 100.00

OlympiadBench

Baseline accuracy: Student 41.69% | Teacher 55.04%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	54.30	53.12	54.60	**55.49**	53.12
2.5	53.12	52.52	54.01	53.86	51.48
3	51.34	51.63	51.48	51.34	50.59
3.5	50.30	50.45	50.74	47.63	47.48
4	49.55	46.14	46.44	48.52	47.48

Color scale: 46.14 – 55.49

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	1.19	4.23	7.58	13.66	23.36
2.5	0.74	2.62	5.16	9.48	16.95
3	0.49	1.83	3.50	6.54	12.09
3.5	0.37	1.30	2.41	5.03	8.86
4	0.25	0.91	1.84	3.45	6.59

Color scale: 0.00 – 100.00

BBH

Baseline accuracy: Student 43.06% | Teacher 79.81%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	79.17	79.72	79.07	**80.00**	77.87
2.5	79.17	79.26	79.26	78.24	77.87
3	79.26	78.43	78.61	78.98	77.04
3.5	77.87	79.17	78.33	78.61	76.39
4	77.69	78.15	77.96	76.76	77.69

Color scale: 76.39 – 80.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	6.91	16.87	28.71	36.55	55.79
2.5	4.65	9.86	22.61	32.70	51.17
3	4.54	12.74	23.14	29.85	44.81
3.5	3.60	10.34	20.92	24.15	39.96
4	2.87	10.34	14.93	26.79	37.67

Color scale: 0.00 – 100.00

CRUXEval-O

Baseline accuracy: Student 21.88% | Teacher 46.75%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	45.50	45.12	44.88	45.00	45.38
2.5	45.50	45.12	45.00	45.12	45.25
3	45.88	45.75	45.12	45.50	45.62
3.5	46.00	45.88	45.38	45.50	45.75
4	**46.12**	46.00	45.25	45.50	45.75

Color scale: 44.88 – 46.12

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	2.41	6.01	12.48	13.85	13.40
2.5	2.36	4.96	9.33	11.08	10.46
3	1.57	7.04	8.06	9.09	10.46
3.5	1.45	5.96	6.20	7.96	8.88
4	1.35	4.83	5.66	8.48	8.43

Color scale: 0.00 – 100.00

CRUXEval-I

Baseline accuracy: Student 19.75% | Teacher 50.75%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	**49.00**	47.62	48.12	48.88	48.88
2.5	48.88	47.38	47.75	48.75	48.75
3	48.25	47.25	47.62	48.50	48.75
3.5	47.12	47.25	48.12	48.75	48.88
4	47.50	46.88	47.88	48.50	48.88

Color scale: 46.88 – 49.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	22.14	48.55	63.22	78.42	56.04
2.5	21.54	48.08	63.04	78.34	40.05
3	20.91	47.61	62.54	77.96	34.69
3.5	20.48	47.13	62.31	77.81	31.26
4	19.83	46.46	61.92	77.58	29.33

Color scale: 0.00 – 100.00

HumanEval+

Baseline accuracy: Student 62.80% | Teacher 87.20%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	81.10	81.71	82.93	**85.98**	84.15
2.5	82.32	82.32	82.32	85.37	84.15
3	80.49	79.88	82.93	84.76	82.32
3.5	82.32	79.27	82.32	83.54	83.54
4	84.76	79.88	82.32	82.32	81.10

Color scale: 79.27 – 85.98

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	3.58	11.41	19.18	29.10	42.57
2.5	3.18	10.63	18.50	28.00	41.26
3	3.09	10.08	17.71	26.71	39.99
3.5	2.86	9.58	16.73	25.45	39.58
4	2.62	8.96	15.60	24.46	37.02

Color scale: 0.00 – 100.00

MBPP+

Baseline accuracy: Student 51.85% | Teacher 72.49%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	71.96	72.49	73.02	72.22	72.22
2.5	73.28	72.75	72.75	72.75	71.96
3	71.96	72.75	71.69	72.22	71.96
3.5	73.02	**73.54**	72.22	73.28	72.49
4	73.28	72.22	72.75	73.02	72.75

Color scale: 71.69 – 73.54

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	6.68	19.16	30.76	43.63	58.01
2.5	6.17	17.81	28.88	41.72	55.95
3	5.49	16.94	27.14	39.76	54.14
3.5	5.12	16.18	25.66	37.86	51.48
4	4.80	15.25	24.42	36.70	50.05

Color scale: 0.00 – 100.00

LiveCodeBench

Baseline accuracy: Student 17.71% | Teacher 30.86%

Accuracy (%) | k →

Threshold	1	4	8	16	32
2	28.57	27.43	29.71	26.29	28.57
2.5	28.57	27.43	27.43	26.86	30.86
3	26.86	27.43	28.00	29.71	31.43
3.5	28.00	26.86	28.57	29.71	29.71
4	28.00	25.71	26.86	30.29	**32.00**

Color scale: 25.71 – 32.00

Teacher token share (%) | k →

Threshold	1	4	8	16	32
2	5.09	15.25	27.02	40.74	56.49
2.5	3.78	12.54	20.46	33.40	50.39
3	3.22	9.78	17.38	28.79	43.00
3.5	2.49	8.94	14.93	24.04	33.64
4	1.87	7.06	12.47	20.25	32.70

Color scale: 0.00 – 100.00

Accuracy, interventions and length: independently scaled per benchmark. Teacher share: shared 0–100% scale. Darker = larger. Bold = best hybrid accuracy.
Teacher share excludes prompts / discarded drafts; it is not compute cost. Stopping rules and baseline generation differ. Full source: source.md.